INCH-POUND

MIL-B-52097B
2 October 1992
SUPERSEDING
MIL-B-52097A(ME)
7 January 1975

MILITARY SPECIFICATION

BARS, CARRYING: MULTIPLE AND SINGLE, M4 BALK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers multiple and single carrying bars for the M4 balk.

1.2 Classification. Carrying bars shall be of the following types, as specified (see 6.2):

Type I - Multiple balk carrying bar.
Type II - Single balk carrying bar.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5420

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SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

FEDERAL

FED-STD-595	- Colors used in Government Procurement.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWINGS

ME

13220E6843	- Bar, Carrying: Multiple Balk.
13220E6844	- Bar, Carrying: Single Balk.

(Copies of drawings required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: SATBE-JBS, Fort Belvoir, VA 22060-5606.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers, and Welding and Brazing Operators.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY, INC. (AWS)

A 5.5	- Low-Alloy Steel Covered Arc-Welding Electrodes.
D 1.1	- Structural Welding Code-Steel.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 N.W. Seventh Street, Miami, FL 33125.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT (UCC)

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT (NMFTA)

National Motor Freight Classification Rules.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 2200 Mill Road, Alexandria, VA 22314.)

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

D 3953	- Strapping, Flat Steel and Seals.
D 4675	- Selection and Use of Flat Strapping Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Description. Carrying bars shall be as shown on the following drawings and as specified herein:

Type I - 13220E6843
Type II - 13220E6844

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the carrying bar. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as shown on the applicable drawings and as specified herein.

3.3.1 Material deterioration prevention and control. The item shall be fabricated from compatible materials, inherently corrosion resistant, or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the item may be exposed.

3.3.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish, or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.3 Welding electrodes. Welding electrodes shall conform to AWS A5.5 classification E7015 or E7016, and size as required.

3.4 Identification marking. The equipment shall be identified in accordance with MIL-STD-130.

3.5 Treatment and painting. The bars shall be cleaned, treated, and painted in accordance with MIL-T-704, type B, except that color designation shall be 34088 in accordance with FED-STD-595.

3.6 Workmanship.

3.6.1 Metal fabrication. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces

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shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.6.2 Welders and welding.

3.6.2.1 Welders or welding operator. Before assigning any welder or welding operator to manual welding work covered by this specification, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed by either AWS D 1.1 or the ASME code for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined. Contractors who only make horizontal welds need not qualify welders for "all position welding". The contractor is responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS and ASME codes. In the event of evidence of poor welds, the Government reserves the right to require retesting of any welder or welding operator (see 6.4).

3.6.2.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of the American Welding Society publication "D1.1-Structural Welding Code-Steel". All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean. Welding procedures shall be qualified in accordance with applicable ASME or AWS code, and all welding and welds shall be in accordance with ASME or AWS code.

3.6.2.3 Welding jigs. Components shall be assembled in steel jigs or frames and welded while securely held in position. The jigs or frames shall be of a design that will limit the distortion to an amount that can be removed without changing the physical characteristics of the metal of the parts being welded.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality

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program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging for delivery (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The preproduction carrying bar shall be examined for defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling for examination. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The carrying bars, as applicable, shall be examined as specified herein for the following defects:

Major

101. Dimensions not in accordance with drawings (see 3.1 and 3.1.1).
102. Material not as specified (see 3.3).
103. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable storage and operating environments (see 3.3.1).

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104. Contractor does not have documentation available for identification of materials, material finishes, or treatment (3.3.2).
105. Incorrect assembly (see 3.1.1).
106. Edges not round (see 3.6.1).
107. Weld defects such as base-metal cracks adjacent to or behind welds, crater cracks or cracks in weld metal, spatter, undercutting of base metal, welding distortions, or hooks not in alignment (see 3.6.2.2 and 3.6.2.3).
108. Welding and welds not as specified (see 3.6.2).
109. Welders and welding operators not certified as specified (see 3.6.2.1).

Minor

201. Identification marking incomplete or illegible (see 3.4).
202. Treatment and painting not as specified (see 3.5).

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

110. Type I bars not bundled and secured as specified for level A or B (see 5.1.1.1 and 5.1.1.2).
111. Strapping not as specified for level A or B (see 5.1.1.1 and 5.1.1.2).
112. Type II bars not packed in boxes as specified for level A or B (see 5.1.1.1 and 5.1.1.2).
113. Blocking and bracing not adequate to prevent movement or damage for level A or B (see 5.1.1.1 and 5.1.1.2).
114. Marking missing, illegible, incorrect, or incomplete (see 5.2).

5. PACKAGING

5.1 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Type I bars. The bars in quantities of four shall be bundled. Each bundle shall be secured with not less than three equally spaced straps

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conforming to ASTM D 3953 and ASTM D 4675, regular duty, zinc coated, size 1/2 inch width and .015 thick. Seals as applicable.

5.1.1.2 Type II bars. The bars shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, grade B, style optional, or PPP-B-621, Class 2, style optional, in quantities not to exceed the weight limitation of the box. The bars shall be blocked and braced within the box to prevent face movement or damage. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.1.2 Level B. Packing shall be as specified in 5.1.1 for level A except boxes for type II bars shall be domestic type or class as applicable.

5.1.3 Level C. The bars shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings. Containers shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking. Bundles and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but not mandatory.)

6.1 Intended use. The Type I carrying bar is issued as part of a sling to lift a group of M4 balk simultaneously in the assembly of fixed and floating bridges. The Type II carrying bar is used to facilitate manual handling of M4 balk in the assembly of fixed and floating bridges. This item is a component of the following Supply Catalogs: SC 5420-97CL-E09, E36, and E45.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Type of bar required (see 1.2).
d. Time frame required for submission of preproduction model (see 3.2).
e. Level of packing required (see 5.1).

6.3 First article. When first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results, and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or

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test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/ provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD Far Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title
3.6.2.1	DI-MISC-80876	Welding Procedure Qualification Test Report.
3.6.2.1	DI-MISC-80875	Welding Procedures.

6.5 Subject term key word listing.

Carrying bar
Lifting bar
Multiple and single carrying bar

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activity:
DLA - CS

Project No. 5420-0196

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-B-52097B	2. DOCUMENT DATE (YYMMDD) 921002

3. DOCUMENT TITLE
Bars. Carrying: Multiple and Single, M4 Balk

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Geraldine F. Harlow	b. TELEPHONE (Include Area Code) (1) Commercial (703) 704-3469 (2) AUTOVON 654-3468
c. ADDRESS (Include Zip Code US Army Belvoir RDE Center ATTN: SATBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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